CarePilot: A Multi-Agent Framework for Long-Horizon Computer Task Automation in Healthcare

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Abstract

*Multimodal agentic pipelines are advancing human–computer interaction by enabling efficient automation of complex real-world tasks, yet current work largely focuses on short-horizon or general-purpose settings, leaving long-horizon, domain-specific automation, especially in healthcare, underexplored. To address this, we introduce **CareFlow**, a human-annotated benchmark of complex, long-horizon workflows across medical annotation tools, DICOM viewers, EHR systems, and laboratory information systems, where existing VLMs struggle with multi-step reasoning. We propose **CarePilot**, an actor–critic multi-agent framework where the Actor leverages tool grounding and dual-memory (long- and short-term) to predict actions from visual and system context, and the Critic evaluates actions, updates memory, and provides corrective feedback. Through iterative agentic simulation, the Actor learns more robust, reasoning-aware behavior. Experiments show that **CarePilot** achieves state-of-the-art performance, outperforming strong closed- and open-source multimodal baselines by **15.26%** and **3.38%** on the benchmark and OOD dataset, respectively. The code and dataset for this project are available at: Carepilot.*

1. Introduction

In today's digital world, software systems powered by large language models (LLMs) and vision language models (VLMs) form the backbone of modern activity, shaping how humans learn, collaborate, analyze data, and create content across domains, interfaces, and tools [1, 4–6, 13–19, 22, 24–26]. Recent advances in large multimodal models (LLMs) have enabled autonomous software agents that can follow high-level natural language instructions to operate real applications, making complex computer use more accessible and

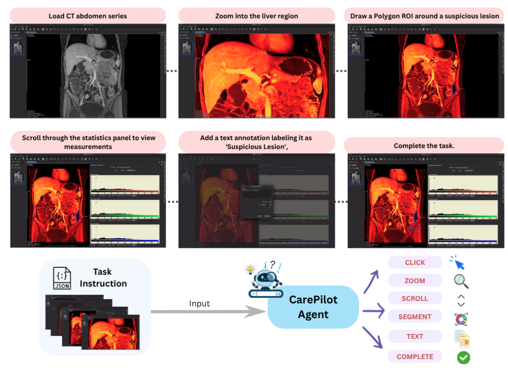

Figure 1. **CareFlow** is a large-scale benchmark for evaluating multimodal agents on a range of real healthcare software. It enables execution-based evaluation and interactive reasoning across DICOM viewers, image annotation tools, EMR/EHR, and LIS platforms. Each task pairs a natural-language goal with GUI screenshots representing authentic clinical workflows.

efficient [47, 49]. However, building agents that can reliably execute long-horizon workflows, spanning dozens of interdependent steps under partial observability, remains a major challenge. A key obstacle is the lack of realistic and interactive benchmarks that capture the heterogeneity of operating systems, interfaces, and domain-specific software environments [34, 46]. Moreover, such long-horizon multimodal agents require robust grounding and memory mechanisms to make informed decisions at each step, a difficult problem that demands effective use of tool representations, contextual reasoning, and long-term memory integration [28, 36, 42, 43].

Healthcare software ecosystems are inherently broad and workflow-centric, spanning DICOM servers/viewers, image-computing and annotation tools, EMR/EHR systems, and laboratory information systems (LIS) [1]. Day-to-day clinical use often requires chaining 10–15 dependent actions, for example, opening a study, configuring views, annotating or

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[1]https://developer.nvidia.com/blog/monai-integrates-advanced-agentic-architectures-to-establish-multimodal-medical-ai-ecosystem/

measuring, exporting artifacts, and updating records, while adhering to data integrity, audit trails, and strict privacy policies. These platforms are highly heterogeneous and policy-constrained, and they evolve frequently: user-interface updates, custom deployments, and institution-specific configurations make agents that overfit to surface layouts brittle. This combination of heterogeneity, long-horizon dependencies, and strict compliance requirements makes healthcare a natural yet uniquely challenging testbed for long-horizon GUI agents [37].

Despite recent progress on long-horizon multimodal agents in Android, desktop, and web environments [35, 40, 46, 48], there remains no standardized public benchmark for healthcare or clinical settings that reflects how users interact with multiple medical softwares. This absence of domain-grounded evaluation makes it difficult to assess how current agents generalize to healthcare-specific tasks typically performed by trained medical professionals [34]. Addressing this gap is essential for developing robust, trustworthy multimodal agents capable of operating safely and efficiently in clinical software ecosystems.

With this motivation, we introduce **CareFlow**, a healthcare-specific long-horizon benchmark that evaluates complex workflows requiring domain knowledge of specialized software. **CareFlow** contains tasks spanning 8–24 consecutive decisions, executed over sequences of GUI screenshots from real medical softwares. At each timestep t, the agent receives the current screenshot, the task instruction, and a condensed history of prior states/actions, and must predict the next semantic action to advance the workflow.

A key challenge in building such benchmarks lies in constructing long-horizon queries that are both high-quality and faithfully reflect real-world software usage. To ensure realism, we collaborated closely with domain experts to draft seed instructions covering the core operations they routinely perform. For each instruction, we recorded detailed step-by-step workflows required to complete the corresponding task. We then filtered and refined these trajectories to retain high-frequency, high-value procedures that are critical in everyday clinical practice. For example, in medical image annotation, we focused on 3D Slicer [12], one of the most widely adopted open-source tools for volumetric analysis, and curated representative workflows for annotation, segmentation, and measurement tasks.

To enable multimodal LLMs to tackle complex, domain-specific, long-horizon workflows in healthcare software ecosystems, we propose ***CarePilot***, a memory- and tool-augmented multi-agent framework inspired by the actor–critic paradigm [20]. At time step t, the **Actor** (a multimodal LM) receives the current screenshot and instruction, invokes lightweight tool modules (e.g., zoom/crop, OCR, UI/object detection) to obtain grounding signals, and predicts the next semantic action. A dual-memory design underpins the system: the *long-term memory* compacts the history up to $t-1$ (key states, actions, outcomes), and the *short-term memory* records the most recent decision and feedback at time t. The **Critic** evaluates the Actor's proposal, updates both memories with observed effects, and issues corrective feedback, during training comparing the Actor's action to reference traces and during evaluation relying on execution outcomes or verifier feedback. If accepted, the action advances the workflow; if revised, the Actor re-plans. At time $t+1$, the Actor conditions on the refreshed memory and grounding signals to produce a more informed action.

Our work makes the following key contributions:

- **Problem Formulation.** We define a new task of *long-horizon computer automation for healthcare software*: given a natural-language goal and a sequence of screenshots, an agent must predict step-by-step actions to complete real clinical workflows.
- **Benchmark.** We present **CareFlow**, an expert-annotated benchmark of long-horizon healthcare workflows comprising 8–24 steps for each task encompassing four major clinical systems. Each task is labeled with interface invariant semantic actions and verified using artifact/state based checks.
- **Framework.** We propose ***CarePilot***, a multi-agent framework built on the actor–critic paradigm that integrates tool grounding with dual memories (long and short-term) for robust next-action prediction.
- **Evaluation.** Extensive experiments across all **CareFlow** domains show that ***CarePilot*** achieves state-of-the-art results, improving task accuracy upto **15.26%** over strong open- and closed-source baselines.

2. Related Work

2.1. Autonomous Multimodal Agents

Recent advances in multimodal agents have enabled models to perceive, reason, and act within digital environments by grounding visual and textual inputs into executable actions. Systems such as Mind2Web [10], SeeAct [49], and UI-TARS [31] leverage screenshot-based reasoning and instructions to automate interactions across web and desktop applications. Large-scale benchmarks including WebArena [50] and AppWorld [41], further extend these capabilities to diverse real-world contexts. However, these efforts primarily target short-horizon, general-purpose tasks where domain-specific reasoning remains limited. To improve temporal coherence and planning, several works have explored memory-augmented and actor–critic-based agents. Voyager [42], Reflexion [36], and Jarvis 1 [43] demonstrate the importance of episodic memory, self-reflection, and long-term credit assignment for persistent task execution. However, the medical domain still lacks agentic systems capable of operating in real-world clinical environments to assist in downstream

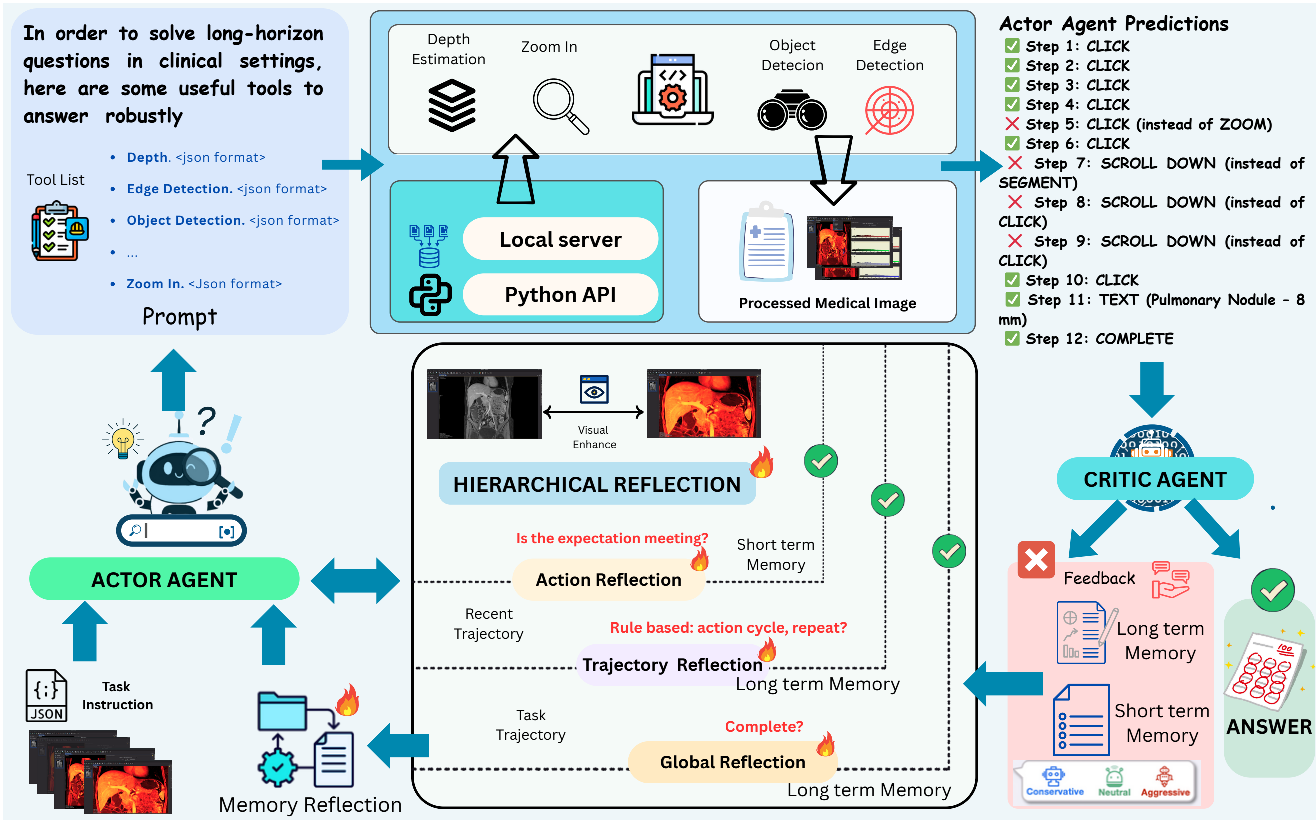

Figure 2. **Overview of the *CarePilot* framework.** An Actor–Critic multi-agent architecture governs hierarchical decision-making for long-horizon healthcare workflows. At each step, the **Actor** observes the current interface and instruction, integrates tool-grounding signals, and its past experience that is stored in short and long-term memories, and predicts the next semantic action. The **Critic** evaluates outcomes, provides corrective feedback, and updates both short and long-term memory buffers to guide subsequent decisions.

tasks such as diagnosis, workflow optimization, and decision support. Existing approaches primarily focus on general or robotic settings, with limited emphasis on clinical reasoning and safety-critical adaptability. Building on these insights, our proposed ***CarePilot*** introduces a dual-memory actor–critic framework that couples long-horizon experience replay with short-term contextual grounding. This design enables robust, reasoning-aware action prediction and adaptive correction across complex, multi-step healthcare workflows.

2.2. Healthcare Software Automation

Automation in healthcare software has traditionally relied on rule-based or heuristic systems for EMR/EHR management, DICOM visualization, and laboratory workflows [27, 32], which lack generalization and multi-stage reasoning. Recent multimodal medical AI focuses on perception tasks like imaging [7] and report generation [21], but not interactive control. **CareFlow** addresses this by introducing a human-annotated benchmark of long-horizon healthcare software interactions across EMR systems, annotation tools, and hospital applications. Combined with ***CarePilot***, it forms the first end-to-end multimodal agentic framework for perception, reasoning, and action in complex healthcare environments, enabling safer and more generalizable automation.

3. CareFlow

To systematically evaluate multimodal LLMs on long-horizon healthcare tasks, we introduce **CareFlow**, a high-quality benchmark of real-world software workflows. This section details the benchmark's composition, statistics, and characteristics, and describes the complete annotation pipeline used to construct **CareFlow** (Figure 2).

3.1. Dataset Pipeline

The **CareFlow** dataset is constructed through a carefully designed four-stage annotation pipeline to ensure diversity, realism, and reproducibility of healthcare workflows.

(i) Crafting Seed Tasks. We collaborated with domain experts to map each software system's real-world usage patterns, functional scope, and operational constraints. Through brainstorming sessions, we identified the core activities performed by practitioners and distilled a seed inventory of

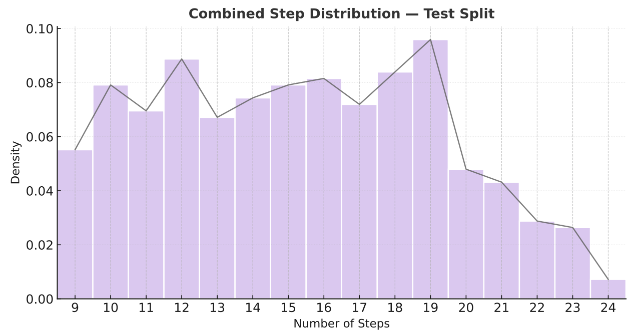

Figure 3. Distribution of task lengths (number of steps) in the test split of ***CareFlow***.

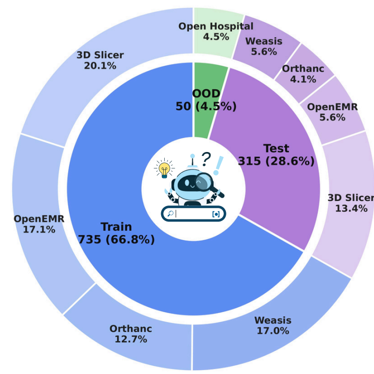

Figure 4. Category distribution of tasks in ***CareFlow*** across four major healthcare software domains.

executable, end-to-end tasks representative of authentic clinical workflows.

(ii) Expanding Diversity and Scale. To broaden coverage and increase sample count, we systematically generated diverse variants of each seed instruction. These variations included controlled substitutions (e.g., replacing "MRI report" with "X-ray report"), parameter adjustments (filenames, thresholds), and procedural edits such as adding or omitting optional zoom or configuration steps, while preserving intent and executability.

(iii) Stepwise Annotation of GUI States. Each generated task was decomposed into a clear sequence of atomic steps by trained annotators. For every step, annotators captured the corresponding screenshot and labeled the precise next semantic action required to progress within the interface. This produced fully grounded, screenshot–action pairs for long-horizon reasoning.

(iv) Quality Assurance and Filtering. We retained only those trajectories that met three strict criteria: (a) chronological consistency of screenshots, (b) task completeness with optimal or near-optimal step sequences, and (c) clear, unambiguous natural-language instructions. Any instance failing one or more of these checks was discarded.

The entire annotation process was supervised by two domain experts who routinely use these healthcare software systems, while two trained interns populated the images and task formulations under their guidance. The test set was independently validated by the experts, and inter-annotator agreement, measured using Cohen's kappa (κ), was 0.78.

3.2. Dataset Characteristics

CareFlow spans four major categories of healthcare software: (i) DICOM viewing and infrastructure (Orthanc, Weasis), (ii) medical image computing and annotation (3D-Slicer), (iii) hospital information and EMR systems (OpenEMR), and (iv) laboratory information systems (OpenHospital). The benchmark contains 1,100 tasks collected across these platforms, each defined by a complex natural-language

Table 1. Core mouse and keyboard actions in ***CarePilot***. Arguments in parentheses denote examples; (x, y) are pixel coordinates, n is a signed step count, and s is a text string.

Function	Description
`CLICK`	Move the cursor (if needed) and click at the specified item in the screen
`SCROLL`	Scroll the active view vertically or horizontally by n units.
`ZOOM`	Adjust the magnification level of the displayed image or view.
`TEXT`	Type string s into the focused input field (e.g., patient ID, study date, filter query).
`SEGMENT`	Create or edit a segmentation or region of interest (ROI) on the displayed medical image.
`COMPLETE`	Mark the workflow or task as finished.

Table 2. Dataset statistics of ***CareFlow***.

Split	# Tasks	Avg. Steps	Min	Max	OOD	Actions
Train	735	12.7	7	22	–	6
Test	315	12.9	9	24	50	6
Total	1050	–	–	–	50	6

instruction and a trajectory of 8–24 consecutive GUI screenshots. Each screenshot corresponds to the application state at time step t within a multi-step workflow. For every state t, we provide an *interface-invariant* next-action label in text, indicating the operation required at $t+1$ to advance toward task completion. The action space of ***CareFlow*** includes six core operations, `CLICK`, `SCROLL`, `ZOOM`, `TEXT`, `SEGMENT` and `COMPLETE` covering the primitive interactions needed for complex healthcare software workflows (see Table 1). Figure 5 illustrates the data composition across the five software categories, while Figure 3 shows the distribution of task lengths. This design aligns with recent multimodal GUI benchmarks like GUIOdyssey [23] while addressing the unique challenges of clinical systems.

4. *CarePilot*

Recent VLMs (GPT-4o, Gemini 2.5 Pro, Qwen VL 3) ground and perceive well in general settings but struggle on real healthcare software, with low task completion despite moderate step-wise accuracy. This limitation motivates ***CarePilot***, a framework that combines multimodal grounding, hierarchical reflection, and dual-memory reasoning to robustly automate complex clinical interfaces. The overall framework is shown in Figure 2.

4.1. Task Definition

Given a goal g illustrated in natural language that requires a sequence of $T \in [t\_low, t\_high]$ steps in a healthcare software environment, the agent observes at each time t the current screenshot x_t and history h_t, and must choose a semantic action $a_t \in \mathcal{A}$ such that the overall sequence completes the task correctly. We formalize this as selecting actions that maximize execution success:

$$\hat{a}_{1:T} = \mathbb{1}\Big[V\big(g, x_{1:T}, a_{1:T}\big) = 1 \Big]. \tag{1}$$

where $V(\cdot)$ is a verifier that returns 1 iff the workflow is successfully completed (i.e., all required artifacts and states are achieved).

4.2. Tool Grounding

To better parse healthcare visual interfaces inspired by [45], we integrate four lightweight perception tools into the rollout and feed their outputs back to the MLLM for grounded next-action prediction: (1) **UI object detection (open-vocabulary)**: given a screenshot $\mathcal{I}_{\text{in}}$ and a text query q (e.g., "MPR," "Export," "Orders"), it returns bounding boxes $\mathcal{B}_{\text{out}}$ over matching widgets; (2) **Zoom/Crop**: from a region $\mathcal{R}$ on $\mathcal{I}_{\text{in}}$, it produces a magnified view $\mathcal{I}_{\text{focus}}$ to inspect small controls; (3) **OCR**: extracts token–box pairs $\mathcal{T}_{\text{out}} = \{(w_i, \mathbf{b}_i)\}$ for labels such as series names, patient fields, order IDs, and LIS codes, disambiguating visually similar elements; and (4) **Template/Icon matching**: given $\mathcal{I}_{\text{in}}$ and a template τ (e.g., measure/save/send-to-PACS), it returns matches $\mathcal{M}_{\text{out}}$ robust to themes, scaling, and locales. These four modules provide the best reliability benefit trade-off among tested toolsets. The outputs of these four perception tools are aggregated into a unified representation denoted as ϕ_t. This tool-grounded feature vector ϕ_t serves as the perceptual grounding signal for subsequent modules, conditioning both memory updates and action prediction.

4.3. Memory Utilization

Long-horizon healthcare workflows require reasoning over both current and past contexts. Building on the perceptual grounding from the tool modules, CarePilot further introduces a dual-memory mechanism to reason over both current and past contexts in long-horizon workflows.

At each step t, the agent updates:

$$\mathcal{M}_t^S = f^S(x_{t-1}, a_{t-1}, r_{t-1}), \tag{2}$$

$$\mathcal{M}_t^L = f^L\big(\mathcal{M}_{t-1}^L, \mathcal{M}_t^S, \phi_t\big), \tag{3}$$

where $\mathcal{M}_t^S$ denotes the *short-term memory* summarizing the most recent context (previous screenshot, executed action, and critic feedback r_{t-1}), and $\mathcal{M}_t^L$ denotes the *long-term memory*, a compact trajectory embedding updated using tool-grounding features ϕ_t. The next action is conditioned on both memories:

$$a_t = \pi_\theta(g, x_t, \mathcal{M}_t^S, \mathcal{M}_t^L), \tag{4}$$

where π_θ is the multimodal policy. This dual-memory mechanism stabilizes long-horizon reasoning, mitigates error accumulation, and preserves semantic consistency across workflows, consistent with prior hierarchical memory agents [20, 39]. The resulting short- and long-term memories are then consumed by the Actor–Critic framework to condition future actions and guide reflection-based updates.

4.4. Actor-Critic Framework

Leveraging both the perceptual grounding from tools and the temporal context maintained in memory, the Actor–Critic framework forms the core decision module of CarePilot. Both the Actor and Critic are instantiated from the same multimodal LLM (Qwen-VL 2.5-7B), differing only in their functional roles i.e., *proposal* versus *evaluation*, and their input conditioning.

At time t, the **Actor** observes $(x_t, g, \phi_t, \mathcal{M}_t^S, \mathcal{M}_t^L)$ and samples a semantic action:

$$a_t \sim \pi_\theta\big(a_t \mid x_t, g, \phi_t, \mathcal{M}_t^S, \mathcal{M}_t^L\big). \tag{5}$$

The **Critic**, parameterized by ϕ, evaluates the proposal via a value function:

$$Q_\phi(x_t, g, a_t, \mathcal{M}_t^S, \mathcal{M}_t^L) \to \hat{r}_t \in [0, 1], \tag{6}$$

where $\hat{r}_t$ estimates the action's correctness. If $\hat{r}_t > \tau$, the Critic approves and updates both memories; otherwise, it issues structured feedback δ_t through hierarchical reflection.

Hierarchical Reflection. If a prediction is incorrect ($\hat{r}_t \leq \tau$), the Critic applies a three-level reflection: (i) the *Action Reflector* compares consecutive states (x_t, x_{t+1}) to detect local grounding or perception errors; (ii) the *Trajectory Reflector* inspects a short window $\{a_{t-k}, \ldots, a_t\}$ to diagnose stalled progress or violated preconditions; and (iii) the *Global Reflector* evaluates the entire trajectory $\{a_1, \ldots, a_t\}$ for goal consistency and decide if the task is completed yet or not. The action reflector is stored in the short-term memory, and the trajectory and global reflector gets stored in

Table 3. Results on **CareFlow**. Columns group *Step-Wise Accuracy (SWA)* and *Task Accuracy (TA)* for each software; the last group reports the overall *Average*. Best results are **bold**, best among baselines are underlined. Green-highlighted rows denote our proposed method.

Model	Weasis		3D Slicer		Orthanc		OpenEMR		Average	
	SWA	TA	SWA	TA	SWA	TA	SWA	TA	SWA	TA
Qwen 2.5 VL 7B	58.64	1.33	61.4	1.66	65.5	2.5	63.20	1.66	57.18	1.78
Qwen 3 VL 8B	65.20	1.33	68.38	3.33	67.71	5.0	66.31	5.0	66.90	3.66
Qwen2.5 VL 32B	79.95	5.13	68.00	1.90	48.62	2.42	41.60	0.32	60.72	2.43
Llama 3.2 11B	86.03	10.26	76.47	4.60	69.58	13.58	62.56	11.47	75.65	9.50
Llama 4 Scout	84.34	10.26	78.47	2.70	<u>88.56</u>	23.90	<u>85.55</u>	21.79	<u>85.65</u>	13.50
Llama 4 Maverick	88.21	18.69	71.55	3.40	84.99	27.99	77.97	25.68	80.53	19.20
Qwen3 VL 235B	83.14	17.69	72.44	5.30	87.48	25.40	84.46	24.52	81.85	19.70
Mistral3.2 VL 24B	88.15	5.13	64.81	0.67	68.44	0.79	61.43	0.00	70.65	1.67
Nemotron 12B VL	86.98	12.82	73.95	5.13	73.56	14.46	66.55	12.36	77.93	10.71
GPT-4o	85.30	20.0	77.5	27.37	88.5	26.67	85.1	27.50	83.13	25.40
GPT-5	<u>88.72</u>	<u>31.25</u>	<u>81.42</u>	<u>37.9</u>	86.92	<u>46.67</u>	83.82	<u>31.25</u>	85.22	<u>36.19</u>
Gemini 2.5 Pro	68.90	3.75	59.70	5.26	71.30	6.66	61.70	6.75	65.15	5.39
CarePilot (Qwen 2.5 VL-7B)	90.38	40.00	82.09	54.75	**93.80**	55.00	**90.18**	**56.70**	88.05	48.90
CarePilot (Qwen 3 VL-8B)	**92.50**	**48.76**	**88.90**	**54.80**	91.80	**56.67**	87.52	46.25	**90.18**	**51.45**

long-term memory. The resulting feedback $\delta_t^{(S)}$ and $\delta_t^{(L)}$ update the corresponding memories:

$$\mathcal{M}_{t+1}^{S} = f^{S}(\mathcal{M}_t^{S}, a_t, \delta_t^{(S)}), \tag{7}$$

$$\mathcal{M}_{t+1}^{L} = f^{L}(\mathcal{M}_t^{L}, \delta_t^{(L)}). \tag{8}$$

This hierarchical update promotes localized correction and long-term stability.

4.5. Training Strategy

After simulating actor-critic trajectories, we distill the Critic's feedback into the Actor following a reasoning distillation paradigm [30, 38], eliminating the need for explicit multi-agent evaluation at inference time. The Actor is fine-tuned *exclusively* on Critic-augmented *successful* trajectories $\{(x_i, g_i, \phi_i, \mathcal{M}_i^S, \mathcal{M}_i^L, a_i^\star)\}_{i=1}^N$, where $a_i^\star$ denotes the Critic-verified and corrected next action. Each training sample also includes associated metadata, the updated memory state, and required tool-grounding information, which together form the Actor's full input context at step $t+1$. Because the Actor is trained only on verified successful trajectories, the feedback signal r_{t-1} is always positive, and training follows a teacher-forced assumption in which all previous steps are assumed correct. This avoids any distribution shift during step-by-step autoregressive inference. The supervised fine-tuning loss is:

$$\mathcal{L}_{\text{SFT}} = -\frac{1}{N}\sum_{i=1}^{N} \log \pi_\theta\Big(a_i^\star \mid x_i, g_i, \phi_i, \mathcal{M}_i^S, \mathcal{M}_i^L\Big). \tag{9}$$

At inference, only π_θ is retained: given the current GUI state, instruction, and memory context, the Actor directly predicts the next semantic action without any Critic involvement. In inference, the distilled Actor approximates the Critic's reasoning, eliminating runtime overhead while preserving performance. This design preserves the Critic's structured reasoning and memory usage within the Actor's parametric knowledge, enabling both faster inference and stronger performance compared to zero-shot and explicit actor-critic feedback loops.

5. Experimental Setup

This section outlines our experimental setup, including implementation details (Sec. 5.1), dataset design (Sec. 5.2), baselines (Sec. 5.3), and evaluation metrics (Sec. 5.4).

Implementation Details. All experiments were conducted on NVIDIA A100 (40 GB) GPUs and Google Colab Pro+ environments, with each model trained for roughly 5–6 hours. Our framework was implemented using PyTorch [29], Hugging Face Transformers [44], and Unsloth[2] for efficient fine-tuning, while baselines were accessed via the DeepInfra API[3]. We used a cosine learning-rate schedule with warmup 100, learning rate 2×10^{-4}, weight decay 0.01, and two training epochs. The sequence length was capped at 768 with gradient_accumulation_steps =32, yielding an effective batch size of $32\times$ the number of GPUs. Mixed precision fp16=True, gradient checkpointing, and clipping max_grad_norm =1.0 were enabled. Checkpoints were saved per epoch, logs every 10 steps, and evaluation was disabled (evaluation_strategy=no) to allow uninterrupted epoch-wise

[2]https://github.com/unslothai/unsloth
[3]https://deepinfra.com/

Table 4. Results on **OOD OpenHospital**. Best is **bold**, best among baselines are underlined. Green rows denote our method (***CarePilot***), gray rows indicate closed-source models, and white rows represent open-source baselines.

Model	SWA	TA
Qwen2.5 VL 32B	71.74	12.72
Llama 3.2 11B	70.76	16.36
Llama 4 Scout	72.20	20.75
Llama 4 Maverick	73.71	27.27
Qwen3 VL 235B	75.18	25.46
Mistral 3.2 VL 24B	69.63	1.82
Nemotron 12B VL	72.90	18.18
Gemini 2.5 Pro	73.90	18.87
GPT-4o	74.63	25.48
GPT-5	**79.70**	34.80
CarePilot (Qwen-VL 2.5-7B)	77.93	36.40
CarePilot (Qwen 3 VL-8B)	79.27	**38.18**

training. We fine tune Qwen vision–language backbones using lightweight LoRA adapters (rank 2, lora_alpha 4, dropout 0.1) applied to attention and MLP projections, with base weights loaded in 4 bit precision for efficiency.

Evaluation Metrics. Performance was measured using two complementary metrics: *Step-Wise Accuracy (SWA)* and *Task Accuracy (TA)*. SWA measures the proportion of correct next-action predictions across all steps and tasks, counting a step as correct only when the predicted action exactly matches the annotated label among available options. TA measures the fraction of tasks for which the model predicted all n required actions correctly in order; a task is marked as successful only under this exact-match condition. Together, SWA reflects the reliability of the fine-grained step, while TA captures the success of the end-to-end workflow.

Baselines. To enable an extensive and fair evaluation, we compare both open source and closed source models. On the open source side, we include Qwen models[8] (Qwen 2.5 VL 32B and Qwen 3 VL 235B), Llama models[3] (Llama 4 Scout and Llama 4 Maverick), as well as Mistral 3.2 VL 24B[33] and Nemotron 12B VL[11]. Among closed source systems, we evaluate GPT 4o[2], GPT 5[4], and Gemini 2.5 Pro[9]. All baselines except the GPT ones are accessed via their respective DeepInfra API deployments and they are evaluated in zero shot setting.

6. Results and Findings

We evaluate ***CarePilot*** against strong *open* and *closed* source multimodal baselines across all healthcare domains, providing per–software class breakdowns to highlight strengths and weaknesses. The results, summarized in Table 3, validate our hypothesis that tool grounding and dual (short- and long-term) memory significantly improve long-horizon performance.

[4]https://platform.openai.com/docs/models

Table 5. Results to show the impact of Critic agent in ***CarePilot*** . WC represents without critic. (WC +TG) represents without critic agent but using tool grounding.

Model	SWA	TA
CarePilot(WC)	65.37	3.75
CarePilot(WC + TG)	72.98	12.5
CarePilot (Qwen-VL 2.5-7B)	**88.05**	**48.90**

6.1. Research Questions

R1) How does *CarePilot* perform compared to baselines? ***CarePilot*** consistently outperforms all open- and closed-source baselines across every healthcare domain. In *task accuracy* (TA), the Qwen 3 VL variant reaches 48.76%, surpassing both its Qwen 2.5 VL counterpart (40.00%) and the strongest baseline, GPT-5 (36.19%). For *step-wise accuracy* (SWA), it achieves 92.50%, exceeding Qwen 2.5 VL (90.38%) and outperforming GPT-5 (85.22%) by more than seven percentage points. These consistent gains highlight the impact of tool grounding, hierarchical feedback, and dual-memory design in improving long-horizon reasoning and interface generalization, as detailed in Table 3.

R2) How is the performance of *CarePilot* on Out of Distribution Healthcare Software benchmark? On the Out of Distribution Open Hospital benchmark as shown in Table 4 , ***CarePilot*** with Qwen 2.5 VL achieves an SWA of 77.93 and a task accuracy of 36.40, demonstrating superior robustness relative to strong open source and closed source models. Among open source baselines, Llama 4 Maverick is the strongest with a task accuracy of 27.27; among closed source models, the best reaches 34.80. Overall, the Qwen 3 VL version of ***CarePilot*** is the strongest performer, underscoring its robustness on OOD tasks.

R3) How do open-source models compare to closed source models on *CareFlow*? We evaluate a broad set of open and closed-source multimodal agents, with overall results summarized in Table 3. Among closed-source systems, GPT models achieve the highest average task accuracy (TA), reaching 36.19%, while Gemini 2.5 Pro performs substantially lower. Within open-source models, the `Llama` family shows the strongest results, particularly `Llama-4 Scout`, which matches or surpasses several closed-source systems. Overall, closed-source GPT models maintain a performance lead among proprietary systems, whereas `Llama` variants dominate the open-source group, highlighting a narrowing gap between the two model classes on ***CareFlow***.

R4) How is the performance of *CarePilot* without the Critic Agent? To understand the contribution of the Critic

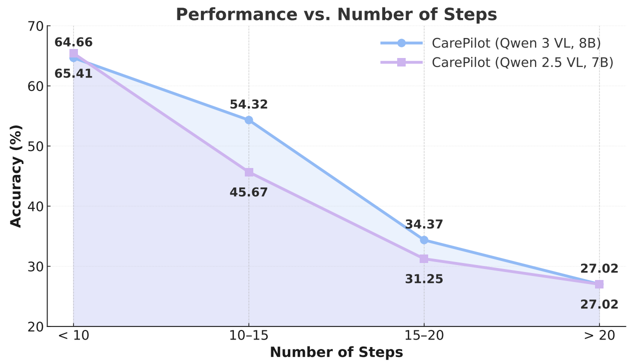

Figure 5. Variation of performance of ***CareFlow*** across different steps range.

Table 6. Ablation on contextual components using ***CarePilot (Qwen 2.5 VL 7B)***: Tool Grounding (TG), Long-Term Memory (LTM), and Short-Term Memory (STM). Reported are overall Step-Wise Accuracy (SWA) and Task Accuracy (TA). Best results are **bold**, and second-best are <u>underlined</u>.

Contextual Information			Overall	
TG	*LTM*	*STM*	**SWA**	**TA**
✗	✓	✓	73.20	9.37
✓	✗	✓	<u>82.10</u>	23.67
✓	✓	✗	80.40	<u>30.42</u>
✓	✓	✓	**88.05**	**48.90**

agent, we conduct experiment as shown in Table 5 where we employ only the Actor agent under two configurations: one with access to tools and one without. In this setting, the task performance (TA) drops to 3.75 without tools and to 12.5 with tools, both of which are substantially lower than the performance of the full ***CarePilot*** framework using the Qwen-2.5 VL 7B model. These findings indicate that the Critic's role in facilitating hierarchical reflection is a key factor underpinning ***CarePilot***'s overall effectiveness.

6.2. Ablation Studies

Impact of each component in *CarePilot*. We conduct extensive ablations in Table 6 to quantify the contribution of each component in ***CarePilot***. Tool grounding (TG) emerges as the most critical: removing TG drops task accuracy to 9.37. This supports our conclusion that proper grounding supplies the context needed for accurate next-action prediction. We also assess the effects of removing Short-Term Memory (STM) and Long-Term Memory (LTM). In our evaluations, LTM proves more consequential; removing it results in a larger performance decline than STM.

Variance of results across different healthcare software. Table 3 shows that 3D Slicer is the most challenging setting (TA $\leq$ 5.3), where ***CarePilot*** achieves the largest gains, highlighting the importance of tool grounding and memory. While Orthanc and OpenEMR have higher baseline TAs (mid-teens to high-20s), ***CarePilot*** still roughly doubles performance, maintaining similar gains on Weasis.

Variation of Performance with Task Length. Figure 5 shows that performance of ***CarePilot*** declines with increasing task length: accuracy remains above 64% for short workflows (<10 steps), drops notably between 10–15 steps (more sharply for the 7B variant), and falls below 35% beyond 15 steps, converging around 27% for tasks >20 steps. Across all ranges, Qwen 3 VL consistently outperforms Qwen 2.5 VL, showing better stability on longer sequences.

Qualitative Analysis. Representative traces show that both **GPT-5** and **LLaMA-4 Maverick** handle setup and completion but fail during mid-sequence *mode switching*, often confusing `CLICK`, `ZOOM`, and `SCROLL`. **GPT-5** is more consistent, while **LLaMA-4 Maverick** fails earlier, especially in segmentation and annotation. In contrast, ***CarePilot*** remains stable through tool grounding and memory feedback, highlighting the importance of contextual reasoning and multi-step consistency for long-horizon tasks. Please check the Appendix for visualisations. **Additional Supplementary Material.** We provide additional details in the supplementary material, including qualitative analyses, the prompts used, further experiments on ***CarePilot*** with the Qwen-3 8B-VL model, extended ablation studies, more information on data sources, inference-performance tradeoff analysis and the ethical considerations followed in this work.

7. Conclusion

We introduce ***CarePilot***, a novel multi-agent framework comprising an action and critic agent for automating long-horizon tasks in healthcare. It combines tool grounding with historical context through two complementary memory modules: a *short-term memory* (STM) that stores the most recent step, outcome, and rationale, and a *long-term memory* (LTM) that maintains trajectory-level context for reasoning across steps. We also present ***CareFlow***, the first benchmark dedicated to long-horizon healthcare software tasks, encompassing samples from multiple platforms across diverse clinical subdomains. Experiments show that ***CarePilot*** achieves state-of-the-art performance, outperforming strong multimodal agents through effective contextual grounding and memory-based reasoning.

Limitation. ***CareFlow*** covers only five healthcare platforms and does not yet capture the full diversity of real-world clinical software. Moreover, ***CarePilot*** predicts high-level semantic actions rather than exact GUI coordinates. Future work will expand platform coverage, add pixel-level grounding, and support longer, multilingual workflows.

8. Acknowledgement

Akash Ghosh would like to sincerely thank MBZUAI for providing the computational resources and infrastructure necessary to conduct the experiments. He also expresses his gratitude to Dr. Muhsin Muhsin, Academic Resident, Department of Community Medicine, IGIMS Patna, and Dr. Maleeka Zainab, Academic Resident, Department of Radiodiagnosis, PMCH, for their valuable guidance in the development of the dataset and validation of the CareFlow benchmark.

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